Data Communications and Networking Fourth Edition

Chapter 18

Virtual-Circuit Networks: Frame Relay and ATM

18-1 FRAME RELAY

Frame Relay is a virtual-circuit wide-area network that was designed in response to demands for a new type of WAN in the late 1980s and early 1990s.

Topics discussed in this section:

Architecture
Frame Relay Layers
Extended Address
FRADs
VOFR
LMI

Frame Relay

Frame Relay

- Prior to FR, they were using a virtual-circuit switching network called X.25.
- Drawbacks of X.25
  - X.25 has a low 64 kbps data rate.
  - X.25 has extensive flow and error control at the data & the network layer

    - Flow & error control at both layers create a large overhead and slow down transmissions.
  - X.25 has its own network layer that user's data are encapsulated in the network layer packets of X.25.

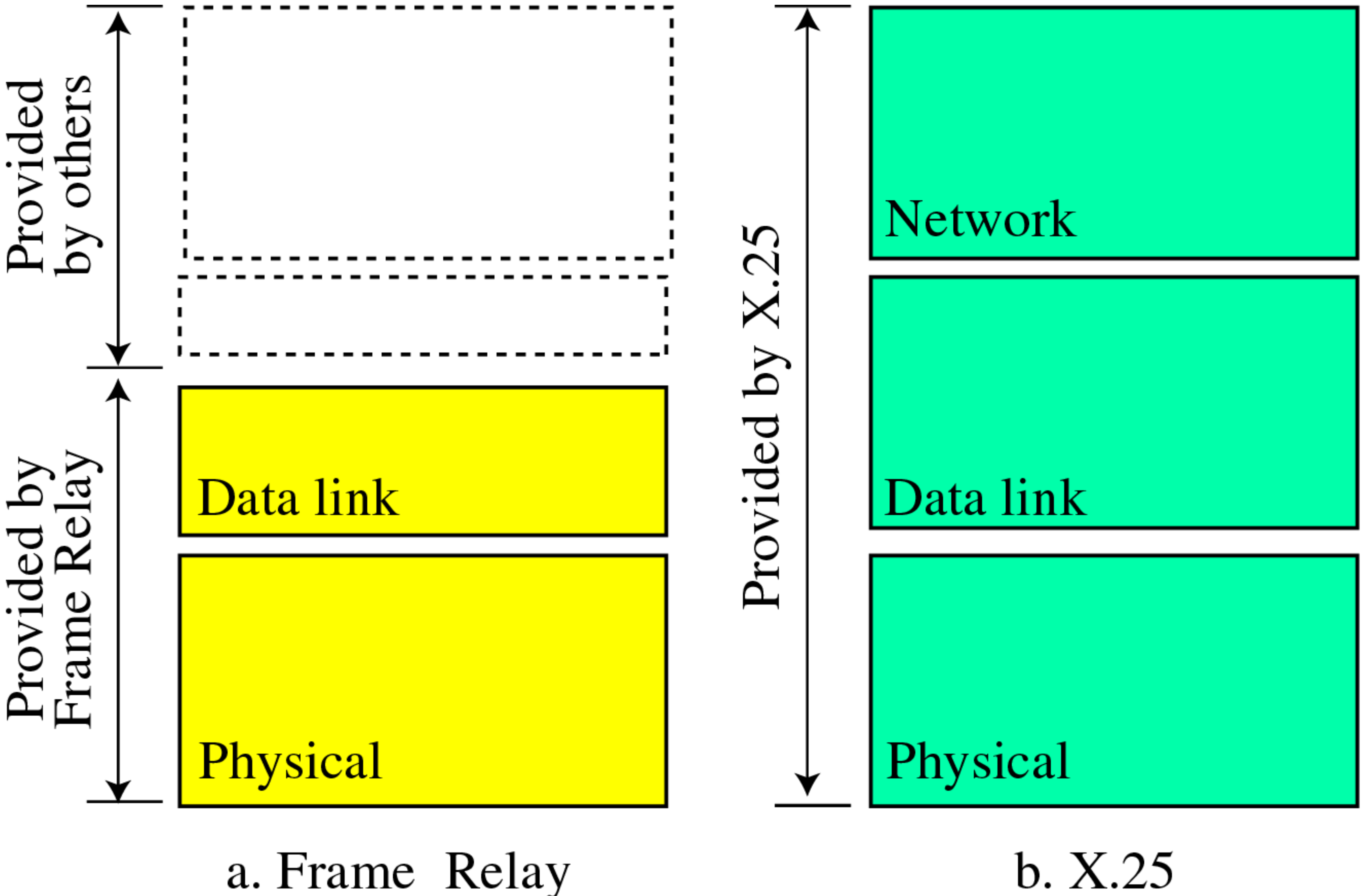
    - Internet has its own network layer.

    - If the Internet wants to use X.25, this doubles the overhead.

X.25 vs. Frame Relay

Feature	X.25	Frame Relay
Connection establishment	At the network layer	None
Hop-by-hop flow control and error control	At the data link layer	None
End-to-end flow control and error control	At the network layer	None
Data rate	Fixed	Bursty
Multiplxing	At the network layer	At the data link layer
Congestion control	Not necessary	Necessary

X.25 vs. Frame Relay

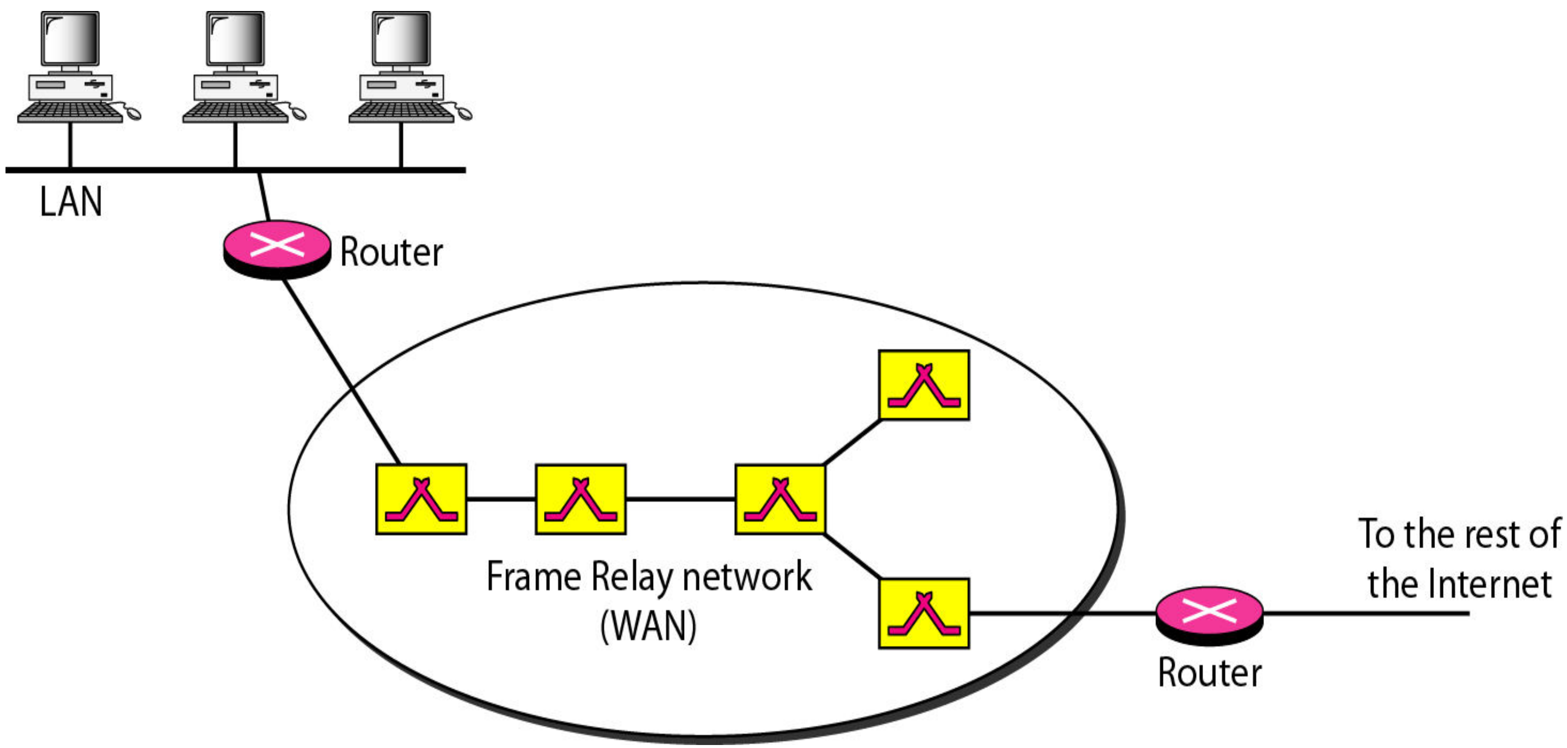


Features of Frame Relay

- Operating in higher speed such as 1.544 Mbps, 45Mbps
- Operating in just the physical and data link layers
- Allowing bursty data
- Allowing a frame size of 9000 bytes, which can accommodate all local area network fame sizes
- less expensive
- Error detection at the data link layer only
  - no flow control or error control

Architecture (Frame Relay)

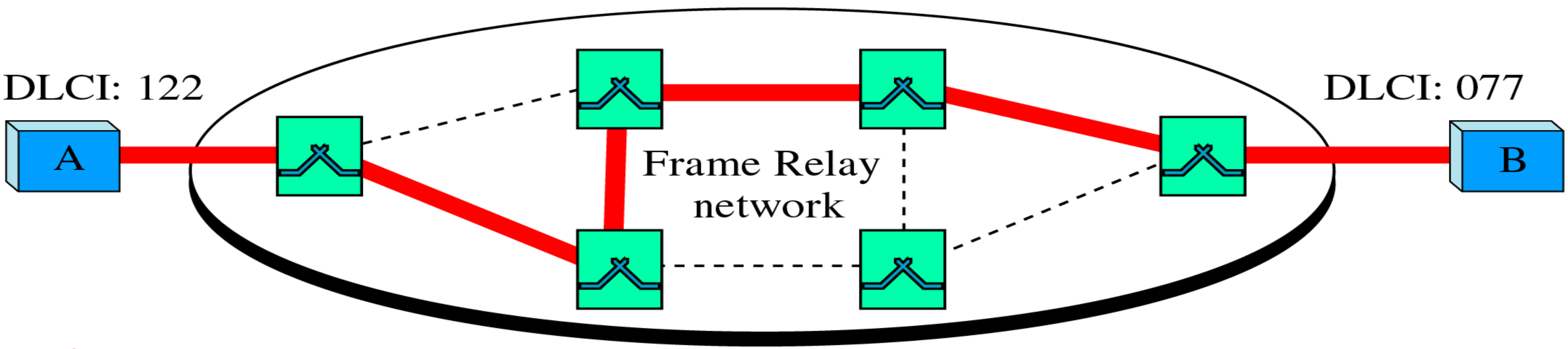
Figure 18.1 *Frame Relay network*



Virtual Circuit (Frame Relay)

- A Virtual Circuit Identifier (VCI) in FR is identified by a number called a Data Link Connection Identifier (DLCI)
- Permanent Virtual Circuits (PVC)
  - The connection setup is simple. The corresponding table entry is recorded for all switches by the administrator.

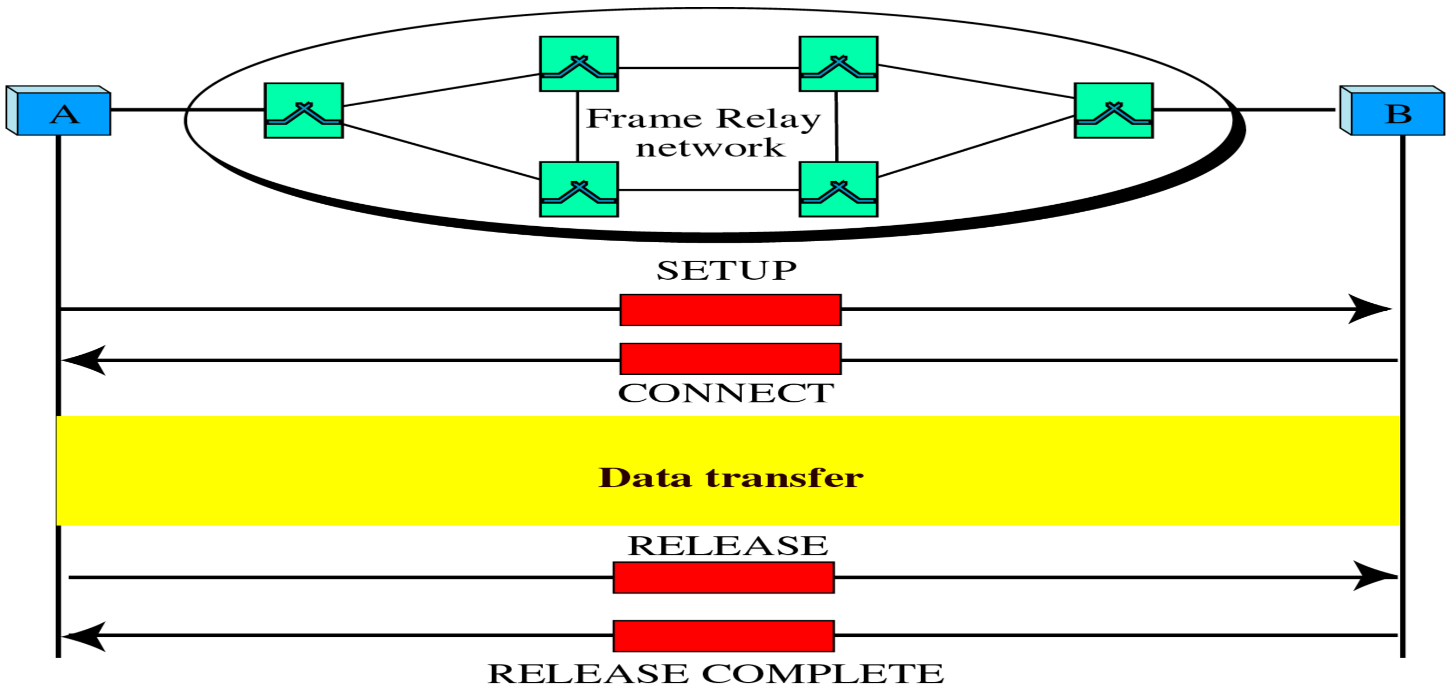
The DLCIs are permanent and assigned by the Frame Relay network provider.



Virtual Circuit (Frame Relay)

Switched Virtual Circuit (SVC)

- The SVC creates a temporary, short connection that exists only when data are being transferred between source and destination.

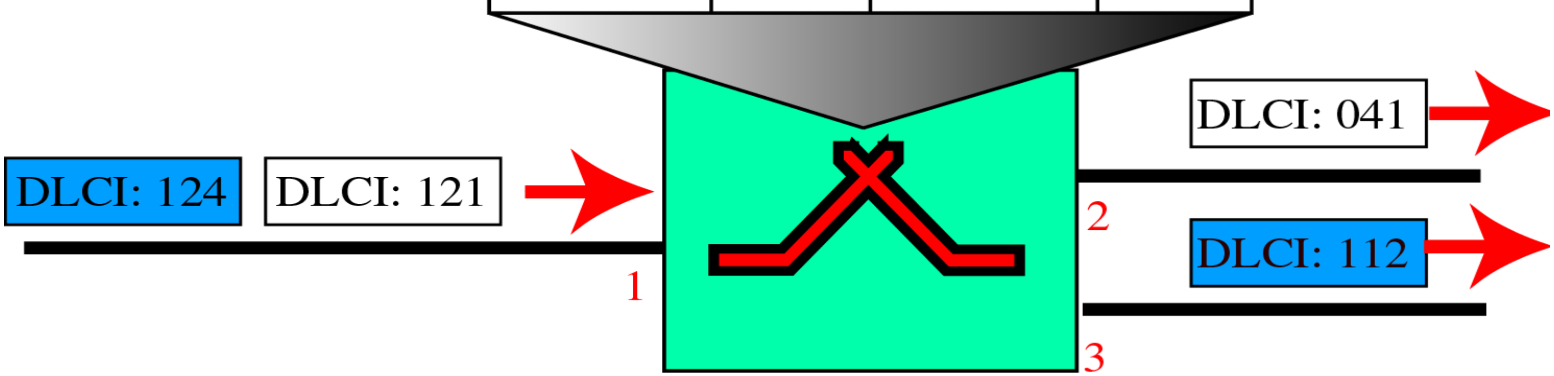


FR Switch

❑ Switches

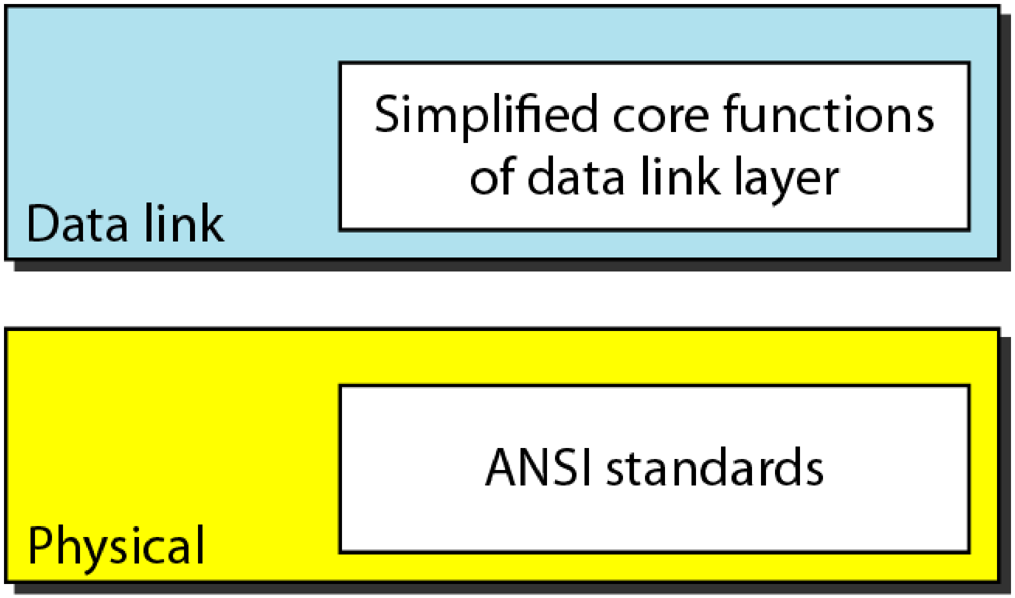
- Each switch in a Frame Relay network has a table to route frames.

Incoming		Outgoing	
Interface	DLCI	Interface	DLCI
1	121	2	041
1	124	3	112
2	167	3	367
3	167	1	192



Frame Relay Layers

- Frame Relay operates only at the physical and data link layers

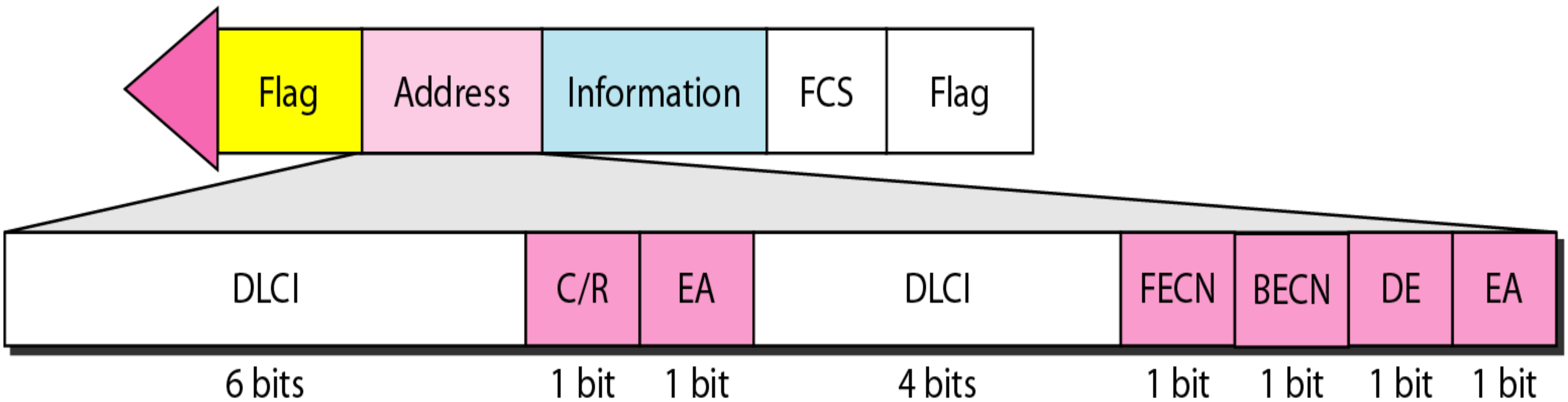


- Frame Relay does not provide flow or error control; they must be provided by the upper-layer protocols.

Frame Relay frame

Figure 18.3 *Frame Relay frame*

C/R: Command/response
EA: Extended address
FECN: Forward explicit congestion notification
BECN: Backward explicit congestion notification
DE: Discard eligibility
DLCI: Data link connection identifier



EA 0 : meaning that another address byte is to follow
DE 1 : to discard this frame if there is congestion

Frame Relay frame

(EXTENDED ADDRESS)

Figure 18.4 *Three address formats*

DLCI			C/R	EA = 0
DLCI	FECN	BECN	DE	EA = 1

a. Two-byte address (10-bit DLCI)

DLCI			C/R	EA = 0
DLCI	FECN	BECN	DE	EA = 0
DLCI			0	EA = 1

b. Three-byte address (16-bit DLCI)

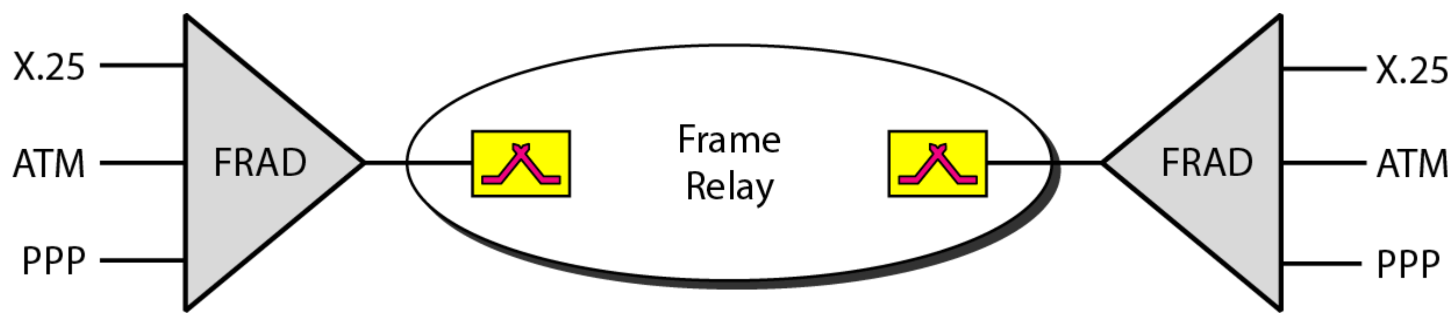
DLCI			C/R	EA = 0
DLCI	FECN	BECN	DE	EA = 0
DLCI				EA = 0
DLCI			0	EA = 1

c. Four-byte address (23-bit DLCI)

FRAD

- FRAD (Frame Relay Assembler/ Disassembler)
  - A FRAD assembles and disassembles frames coming from other protocols to allow them to be carried by Frame Relay frames.

Figure 18.5 *FRAD*

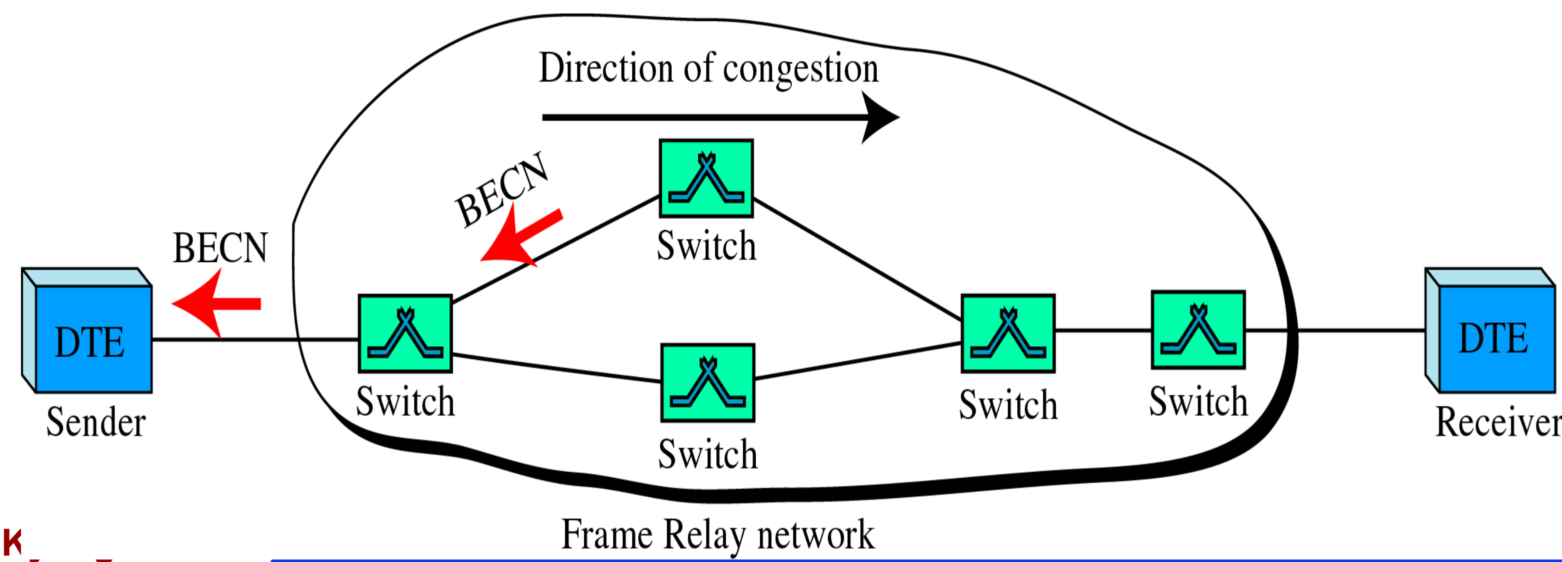


Voice Over Frame Relay

- VOFR (Voice Over Frame Relay)
  - VOFR sends voice through the network.
  - Voice is digitized using PCM and then compressed.
  - The result is sent as data frames over the network.
- LMI (Local Management Information)
  - F/R was originally designed to provide PVC connections.
    - There was not a provision for controlling or managing intefaces.
  - LMI is a protocol added recently to the FR protocol to provide more management features.
    - A keep alive mechanism to check if data are flowing.
    - Multicasting mechanism
    - A mechanism to allow an end system to check the status of a switch.

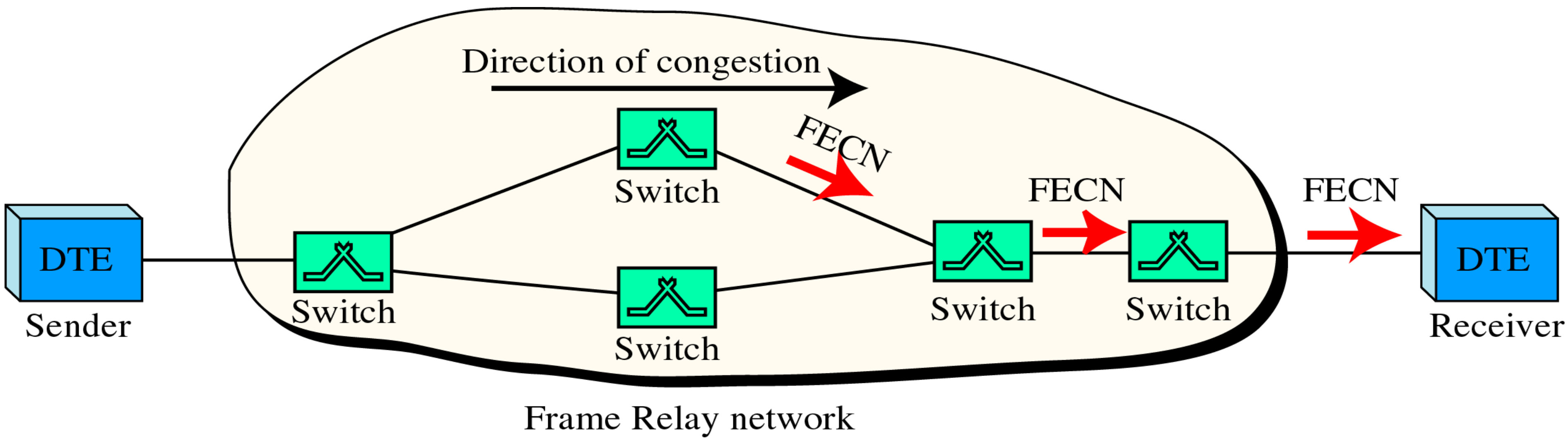
Congestion Control (Frame Relay)

- Congestion Avoidance
  - The FR protocol uses 2 bits in the frame to explicitly warn the source and the destination of the presence of congestion.
- BECN(Backward Explicit Congestion Notification)
  - BECN bit warns the sender of congestion in the network.



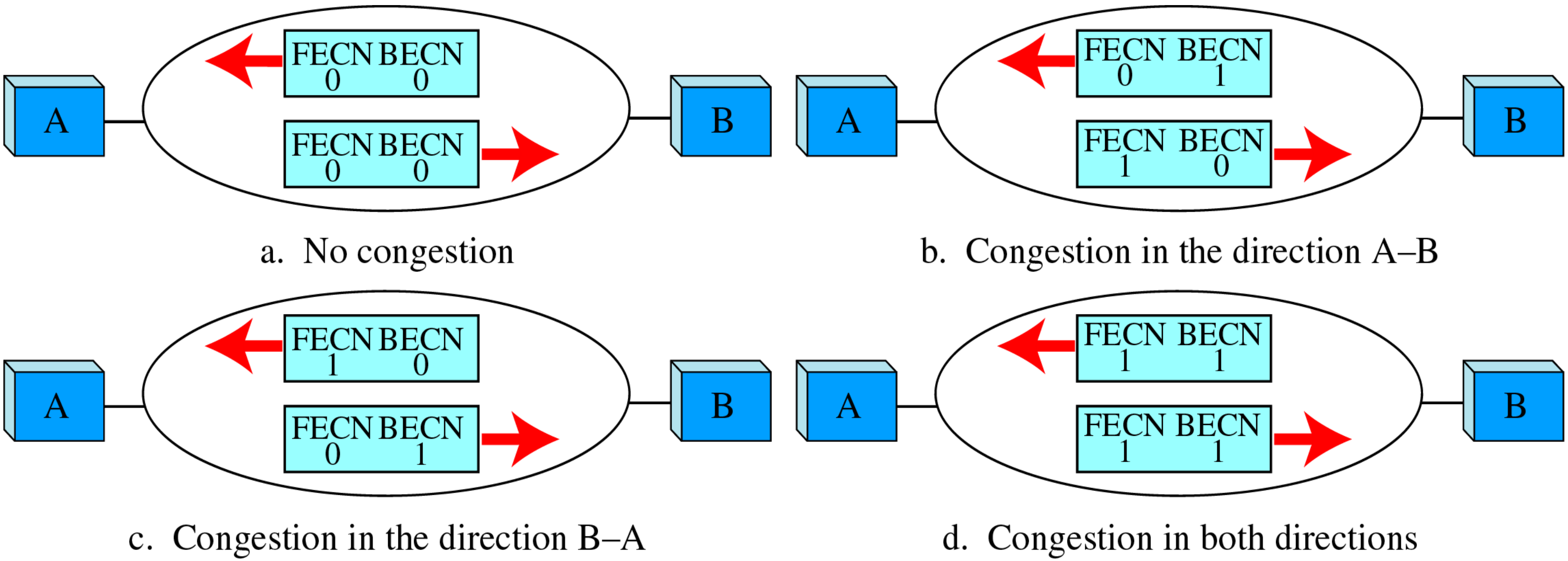
Congestion Control (Frame Relay)

- FECN(Forward Explicit Congestion Notification)
  - FECN bit is used to warn the receiver of congestion in the network.



Congestion Control (Frame Relay)

- 4 cases of congestion



18-2 ATM

Asynchronous Transfer Mode (ATM) is the cell relay protocol designed by the ATM Forum and adopted by the ITU-T.

Topics discussed in this section:

Design Goals
Problems
Architecture
Switching
ATM Layers

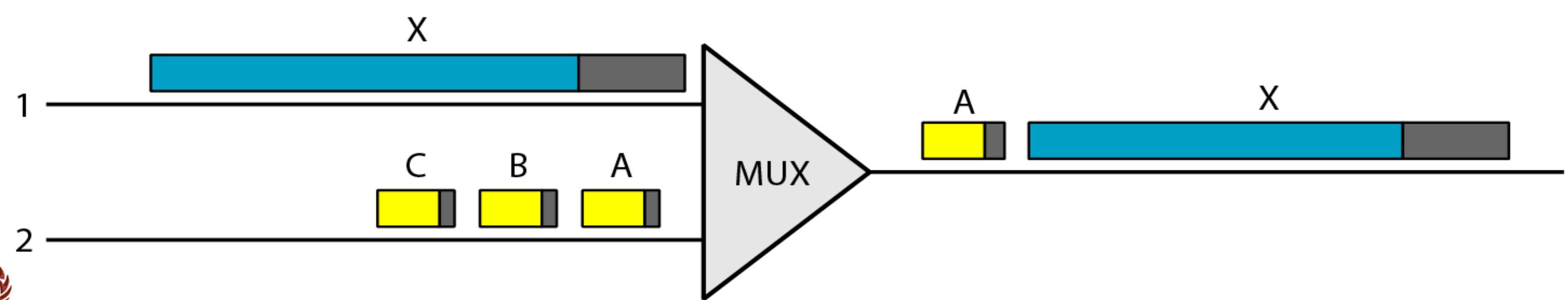
ATM - Design Requirements

- Foremost is the need for a transmission system to optimize the use of high-data-rate transmission media, in particular optical fiber.
  - A technology is needed to take advantage of large bandwidth and strength to noise degradation.
- The system must interface with existing systems.
- Must be implemented inexpensively.
- The new system must be able to work with and support the existing telecommunications hierarchies.
- The new system must be connection-oriented to endure accurate and predictable delivery.
- One objective is to move as many of the function to hardware as possible and eliminate as many software functions as possible.

Problems (ATM)

- Problems associated with existing systems.
  - Frame networks
    - Different protocols use frames of varying size and intricacy.
    - As networks become more complex, the information that must be carried in the header becomes more extensive.
  - Mixed network traffic
    - The variety of frame size makes traffic unpredictable.
    - Internetworking among the different frame networks is slow and expensive at best, and impossible at worst.

    - The sheer size of X creates an unfair delay for frame A.



Cell Networks (Multiplexing Using Cells)

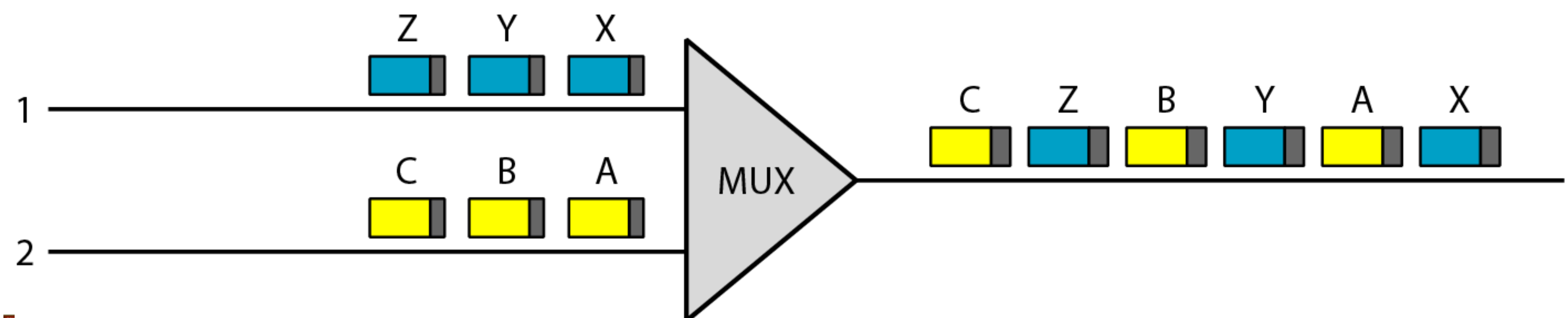
❑ Cell Networks

❖A cell is a small data unit of fixed size.

❖ Cell network uses the cell as the basic unit of data exchange, all data are loaded into identical cells that can be transmitted with complete predictability and uniformity.

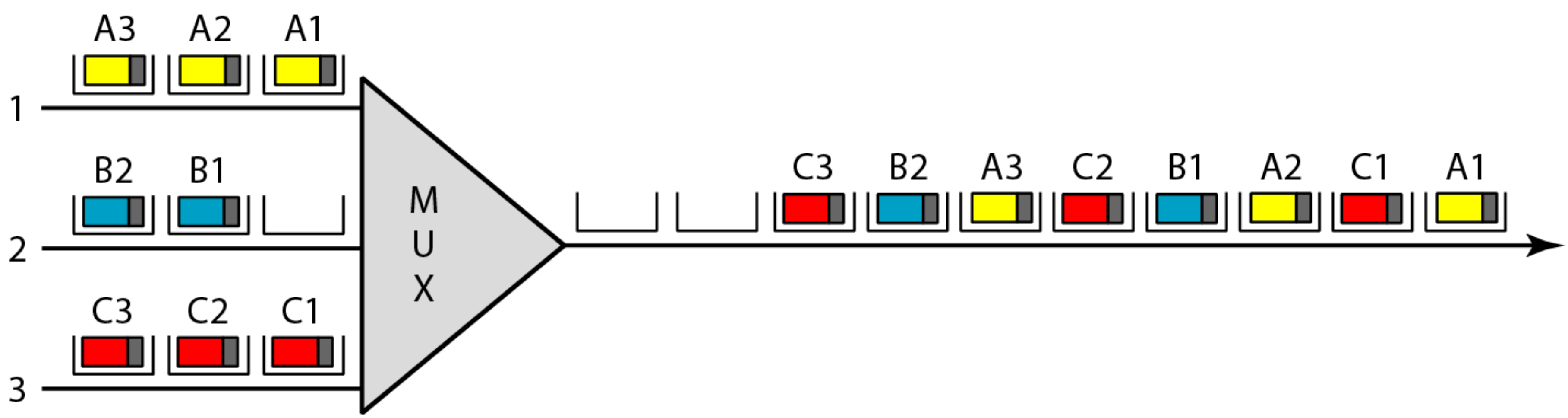
❖Because each cell is the same size and all are small, the problems associated with multiplexing different-sized frames are avoided.

❖Despite interleaving, a cell network can handle real-time transmissions, such as a phone call, without the parties being aware of the segmentation or multiplexing at all.



Asynchronous TDM

- AsychronousTDM
  - ATM uses asynchronous time-division multiplexing – that is why it is called Asynchronous Transfer Mode – to multiplex cells coming from different channels.



Architecture of ATM Network

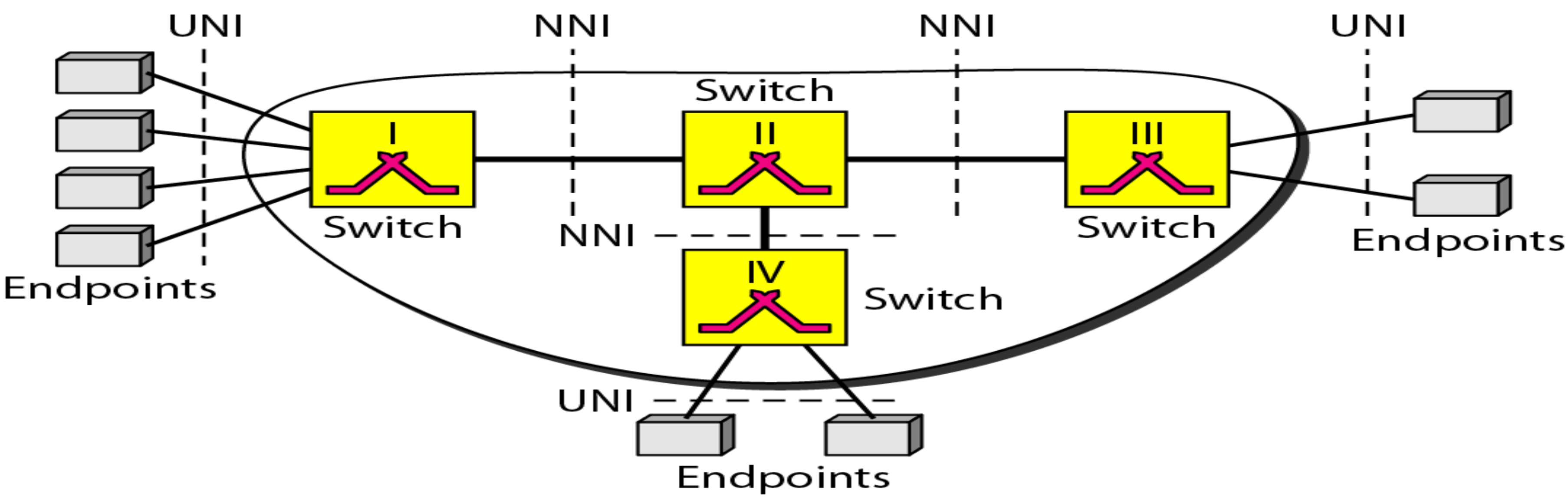
- Architecture

  - ATM is a cell-switched network.

  - The user access devices, called the endpoints, are connected through a UNI (User- to-Network Interface) to the switches inside the network.

  - The switches are connected through NNI (Network-

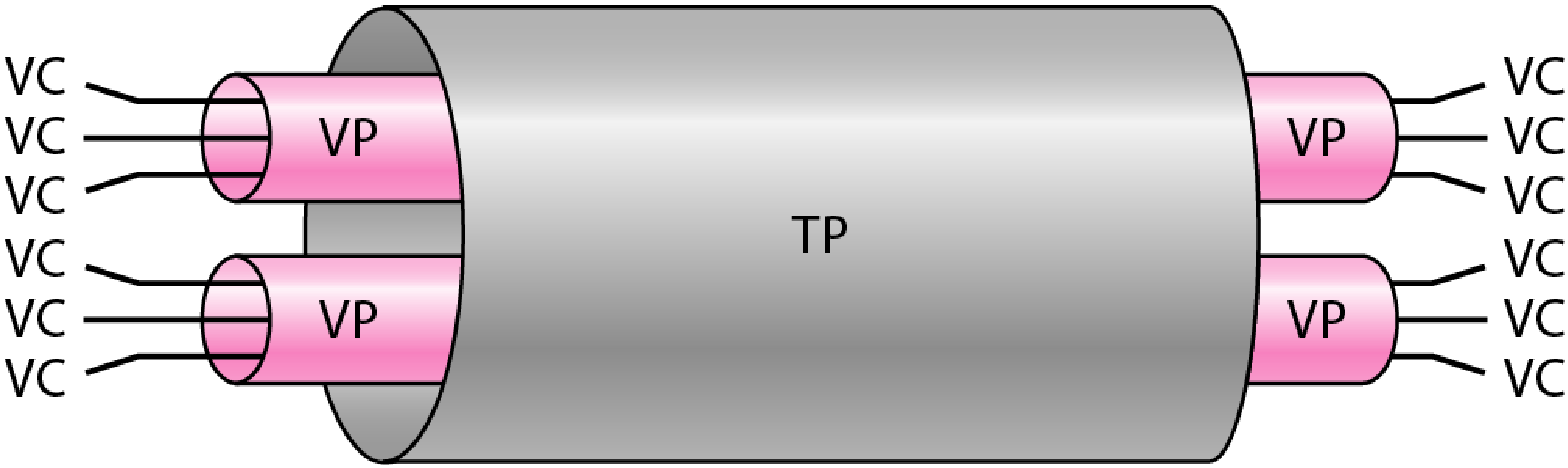
    to-Network interface) .



Virtual Connection of ATM Network

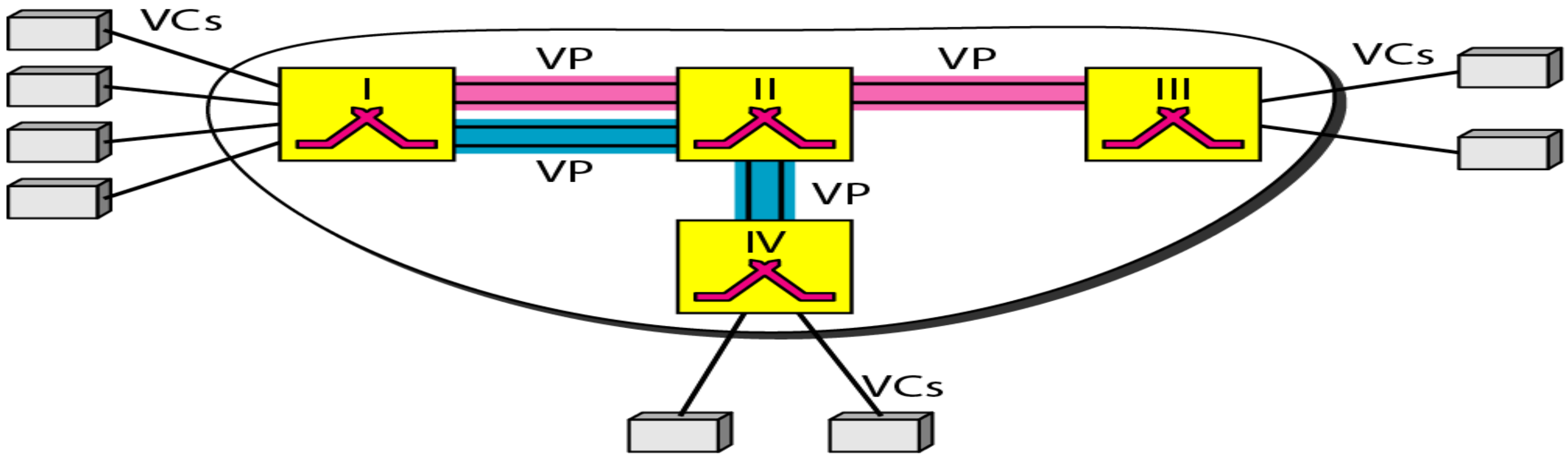
- **Virtual Connection –** Connection between two endpoints is accomplished through transmission paths (TPs), Virtual paths (VPs), and Virtual circuits (VCs).
  - **TP (Transmission Path)**
    - TP is the physical connection (cable, satellite, and so on) between an endpoint and a switch or between two switches.
  - **VP (Virtual Path)**
    - VP provides a connection or a set of connections between two switches. (A TP is divided into several VPs)
  - **VC (Virtual Circuit)**
    - A VP is logically divided into several VCs.
    - Think of a VC as the lanes of highway (VP).

Virtual Connection of ATM

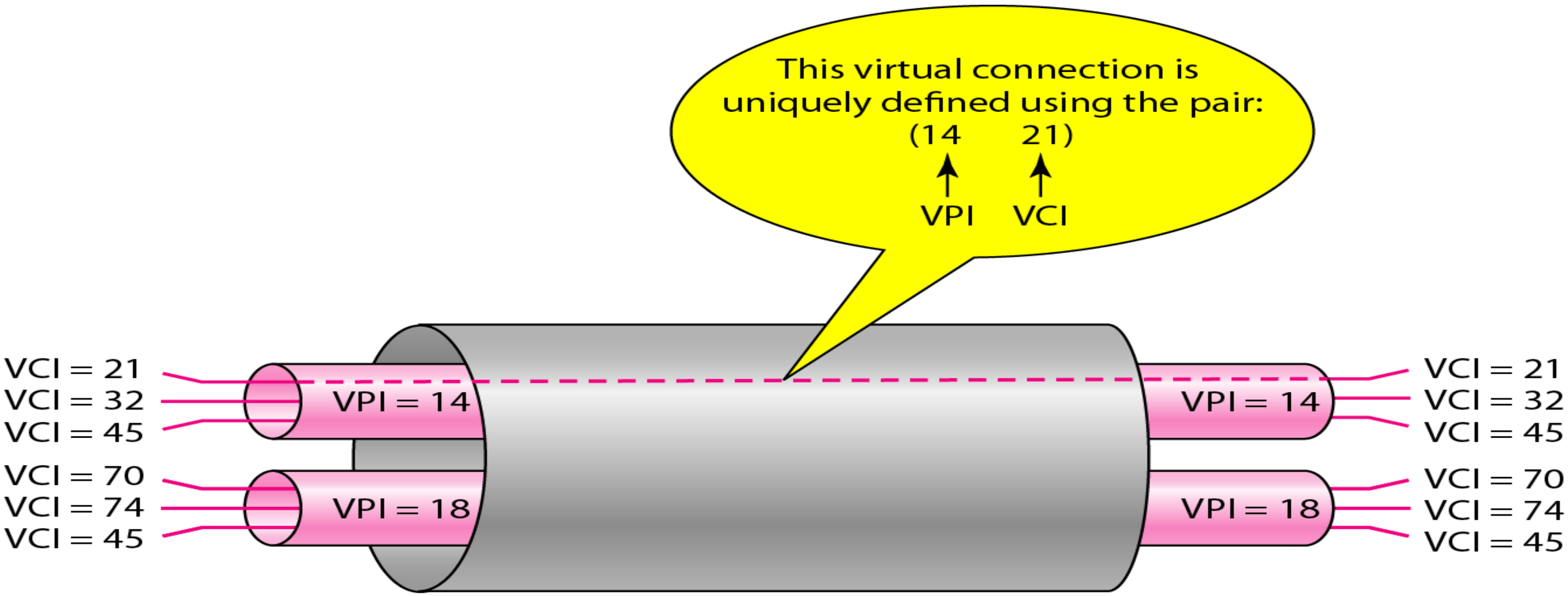


Example of VPs and VCs

- Identifiers
  - VPI (Virtual Path Identifier)
    - The VPI defines the specific VP.
  - VCI (Virtual Circuit Identifier)
    - The VCI defines a particular VC inside the VP.



Connection Identifiers

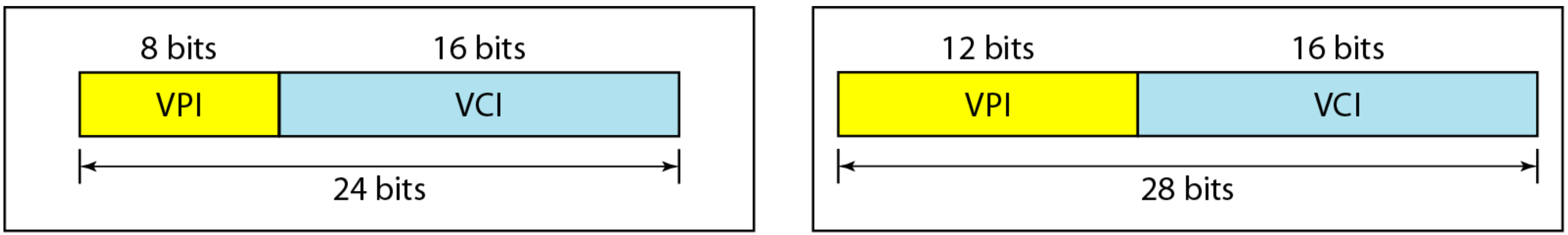


Note that a virtual connection is defined by a pair of numbers: the VPI and the VCI.

Virtual connection identifiers in UNIs and NNIs

VPI for UNI and NNI

- The lengths of the VPIs for UNIs(8b) and NNIs(12b) are different.
- The lengths of the VCI is the same in both interface (16bits)
- Dividing a VCI into two parts is to allow hierarchical routing.



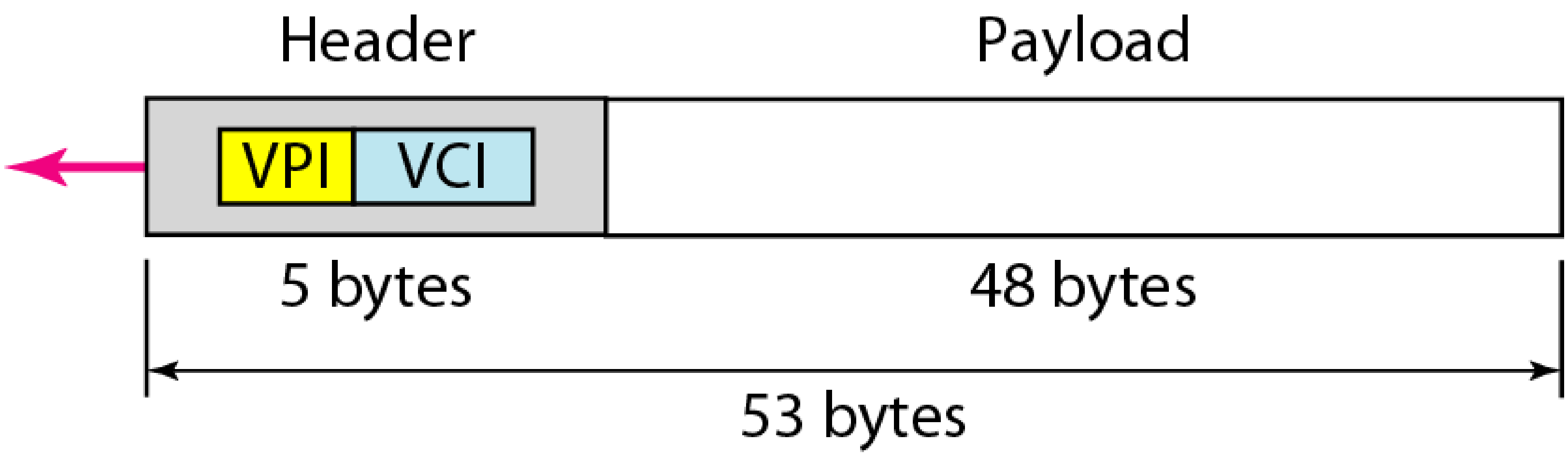
a. VPI and VCI in a UNI

b. VPI and VCI in an NNI

- Most of the switches in typical ATM network are routed using VPIs.
- The switches at the boundaries of the network, those that interact directly with the endpoint devices, use both VPIs and VCIs.

An ATM Cell

- **cells**
  - The basic data unit in an ATM network is called a cell.
  - A cell is only 53 bytes long
    - 5 bytes allocated to the header
    - 48 bytes carrying the payload.



PVC & SVC

Connection Establishment and Release

PVC

- The VPIs and VCIs are defined for the permanent connections, and the values are entered for the tables of each switch.

SVC

- Each time an endpoint wants to make a connection with another endpoint, a new virtual circuit must be established.
- ATM cannot do the job by itself, but needs the network layer addresses and the services of another protocol (such as IP).

Summary (1)

- Virtual-circuit switching is a data link layer technology in which links are shared.
- A virtual circuit identifier (VCI) identifies a frame between two switches.
- Frame Relay is relatively high speed, cost-effective technology that can handle bursty data.
- Both PVC and SVC connections are used in Frame Relay.
- The data link connections identifier (DLCI) identifies a virtual circuit in Frame Relay.
- Asynchronous Transfer Mode (ATM) is a cell relay protocol that, in combination with SONET, allows high-speed connections.
- A cell is a small, fixed-size block of information.
- The ATM data packet is a cell composed of 53 bytes (5 bytes of header and 48 bytes of payload).

Summary (2)

- ATM eliminates the varying delay times associated with different-size packets.
- ATM can handle real-time transmission.
- A user-to-network interface (UNI) is the interface between a user and an ATM switch.
- A network-to-network interface (NNI) is the interface between two ATM switches.
- In ATM, connection between two endpoints is accomplished through transmission paths (TPs), virtual paths (VPs), and virtual circuits (VCs).